PROJECT 3

AWS Lex Quiz Chatbot

Building an Interactive VPC Networking Quiz using Amazon Lex · AWS Lambda · Amazon S3

Praesignis AWS re/Start | Presented by Stacey

CloudLearners Inc.

Project Overview

WHAT & WHY

In this project, we built a fully interactive AWS Networking knowledge-check chatbot for CloudLearners Inc. — an educational startup that needed an engaging, automated quiz experience for its students.

🤖 Part 1 — Simple Bot

Build a basic Lex bot with one intent and one utterance to understand the fundamentals of chatbot creation.

🧠 Part 2 — Quiz Bot

Create a full 15-question VPC networking quiz with branching logic, scoring, and feedback for correct and incorrect answers.

🎤 Part 2 — Presentation

Deliver a live demo and professional presentation of the chatbot to CloudLearners Inc. as the client.

What is Amazon Lex?

AWS AI SERVICE

Amazon Lex is a fully managed AI service from AWS that enables developers to build conversational interfaces (chatbots) using the same deep learning technology that powers Amazon Alexa.

💡 NLP-Powered

Lex uses Natural Language Processing to understand what users type — even when phrased differently from the training utterances.

Use cases: Customer service bots · Knowledge quizzes · Virtual assistants · Booking systems

Intents

What the user wants to do (e.g. StartQuiz)

Utterances

Phrases that trigger an intent (e.g. 'start quiz')

Slots

Data collected from the user (e.g. answer A/B/C/D)

Fulfillment

Lambda function called to process the response

Solution Architecture

HOW IT WORKS

User

Types in the chat window

Amazon Lex

Identifies intent & slot

AWS Lambda

Processes answer & logic

Amazon S3

Stores quiz questions JSON

1. User types 'start quiz' or an answer into the Lex test window or integrated app.
2. Lex matches the input to an intent (StartQuiz or VPCQuiz) using NLP and utterance matching.
3. Lambda is invoked via fulfillment. It reads session state and fetches questions from S3.
4. Lambda evaluates the answer, updates the score in session attributes, and returns the next question.

Client Requirements

CLOUDLEARNERS INC.

CloudLearners Inc. approached us needing an automated, engaging knowledge-check experience for their AWS learners — without requiring a human facilitator.

✅ Interactive quiz chatbot

Must feel conversational, not like a static form

✅ Live running score

Students see their score update after every answer

✅ 15-question VPC quiz

Mix of Beginner and Intermediate difficulty levels

✅ Final grade & encouragement

Personalised completion message based on final score percentage

✅ Clear answer feedback

Correct/incorrect response with an explanation after each question

✅ Scalable question bank

Questions stored in S3 JSON — easy to update without code changes

Part 1 — Building the Simple Bot

BEGINNER EXERCISE

Before tackling the full quiz, we first built a simple one-intent bot (VPCInfoBot) to understand the core Lex workflow.

1

Created the Lex Bot

Named VPCInfoBot · Language: English (ZA) · No advanced config

2

Added VPCInfo Intent

One intent to answer 'What is a VPC?' and similar questions

3

Added Sample Utterances

"What is a VPC?" · "Tell me about VPC" · "Explain VPC to me"

4

Set Closing Response

Configured a plain text explanation of what a VPC is as the bot's reply

5

Built & Tested

Used the Lex test window to confirm the bot responded correctly to utterances

💡 Key Learning: Utterances are the phrases that trigger intents. The more varied your utterances, the better Lex recognises different user inputs.

Technical Approach — Part 2 Quiz Bot

Amazon S3

- Created bucket: netquiz-bot-stacey
- Uploaded quiz_questions.json
- 15 questions with options, answers & explanations
- Lambda reads file on every invocation
- Easy to update — no code changes needed

AWS Lambda

- Runtime: Python 3.12
- IAM Role: S3ReadOnly + BasicExecution
- Handles: StartQuiz, VPCQuiz, FallbackIntent
- Tracks score in session attributes
- Returns formatted question + running score

Amazon Lex

- Bot: VPCQuizBot · Language: English (ZA)
- Intent 1: StartQuiz (no slots)
- Intent 2: VPCQuiz + UserAnswer slot
- Slot type: QuizAnswerType (A/B/C/D)
- Lambda connected via TestBotAlias

Challenges & How We Solved Them

TROUBLESHOOTING

⚠ Invalid Lambda Response: intent name VPCQuiz not valid

Cause: StartQuiz block was returning VPCQuiz as the intent name — Lex rejected the mismatch

✓ **Fix: Changed StartQuiz handler to close() under 'StartQuiz' intent name, not VPCQuiz**

⚠ close() function missing intent_name argument

Cause: The original close() had a hardcoded 'VPCQuiz' string instead of the dynamic intent

✓ **Fix: Added intent_name as a parameter and passed it dynamically in every close() call**

⚠ FallbackIntent is fulfilled — answers not recognised

Cause: Lex could not reliably match single letters (A/B/C/D) to the VPCQuiz intent

✓ **Fix: Added a FallbackIntent handler in Lambda that reads inputTranscript and processes the answer directly**

Quiz Features & Scoring System

🎯 Quiz Features

- 15 questions across Beginner & Intermediate levels
- Question counter shown on every question (e.g. Question 3 of 15)
- ✓ CORRECT! or ✗ Incorrect feedback after each answer
- Full explanation provided for every question
- Running score displayed after each answer (e.g. Score so far: 4/5)
- Final score and personalised grade message at completion
- FallbackIntent catches unmatched inputs and keeps quiz flowing

🏆 Grade Scale

15 / 15 (100%)

Outstanding! Perfect score!

12 – 14 (80%+)

Great job! Strong understanding.

9 – 11 (60%+)

Good effort! Review missed Qs.

Below 9 (<60%)

Keep studying! Revisit docs.

LIVE DEMO

Switching to Amazon Lex Console

- Type 'start quiz' to begin
- Answer Q1 with A, B, C, or D
- See live score update
- Continue to quiz completion

Key Learnings

🔗 Service Integration

Lex, Lambda and S3 must be wired together correctly. Each service has its own permissions model — IAM roles are critical.

🏷️ Intent Name Matching

Lex strictly validates that Lambda's response references the same intent that triggered it. Mismatches cause immediate errors.

📝 Session State

Lambda uses session attributes to remember the question index and score across multiple turns in a single conversation.

🛡️ FallbackIntent as a Safety Net

Single letters like A/B/C/D can confuse Lex's NLP. Handling FallbackIntent in Lambda ensures the quiz never breaks mid-flow.

🗂️ Separating Data from Logic

Storing questions in S3 JSON means quiz content can be updated without touching Lambda code — a scalable, production-ready pattern.

🎤 Communication Skills

Presenting technical work to a non-technical client requires clear language, a structured demo, and confidence in explaining trade-offs.

Thank You

Project 3 — AWS Lex Quiz Chatbot
Praesignis AWS re/Start Program

- ✓ Amazon Lex — Conversational AI interface
- ✓ AWS Lambda — Quiz logic & scoring engine
- ✓ Amazon S3 — Scalable question bank
- ✓ **15-question VPC Networking quiz delivered**

Questions & Discussion